GRAPHFEEDBACK: Score-Guided Black-Box Textual Robustness Evaluation of GraphCLIP

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***Abstract*—Text-attributed graph foundation models expose a robustness surface that is absent from conventional structure-only evaluations: small natural-language edits to an existing node can change its representation while graph topology remains fixed. This paper presents GRAPHFEEDBACK, a bounded two-round black-box procedure that uses observed GraphCLIP class scores to refine graph-aware textual candidates. The method is evaluated on a released GraphCLIP checkpoint and CiteSeer under a six-query budget, a 20% aligned-token edit limit, and a minimum sentence-embedding similarity of 0.85. On a 30-node stratified low-margin stress sample, GRAPHFEEDBACK changes 7/30 initially correct predictions, compared with 5/30 for its matched one-round graph-aware baseline; two successes arise only in feedback-guided refinement, although the paired exact comparison is not significant ($p = 0.5$). A frozen, non-overlapping 60-node class-balanced random experiment does not reproduce this effect: both graph-aware methods achieve 0/60. The evidence therefore supports local feasibility near selected decision boundaries, not population-level superiority. The main contribution is a reproducible, query-bounded methodology that separates attack mechanics, candidate validity, and sampling sensitivity when auditing text-attributed graph models.**

***Index Terms*—text-attributed graphs, GraphCLIP, adversarial robustness, black-box evaluation, score feedback, query efficiency**

I. INTRODUCTION

Graph foundation models increasingly combine structural context with natural-language node attributes. GraphCLIP aligns graph summaries and structural representations to support zero- and few-shot transfer on text-attributed graphs (TAGs) [1]. This capability also creates a distinct robustness question: can a topology-preserving edit to one node's text change its zero-shot prediction? Classical graph attacks perturb edges or node features [2], [3], while recent text-level graph injection work studies the addition of textual nodes [4]. Neither setting isolates the sensitivity of an existing node's text under fixed topology.

Textual attacks traditionally search over discrete substitutions while constraining semantic drift. TextFooler and BERT-Attack establish strong word-level generation baselines [5], [6]; PromptAttack instead asks a language model to generate and refine adversarial text through instructions [7]. Black-box evaluation further requires explicit query accounting because effective search can otherwise depend on thousands of victim interactions [8]. These ideas motivate a small-budget, closed-loop audit for GraphCLIP.

We introduce GRAPHFEEDBACK, a score-only, two-round textual search procedure. It first generates graph-aware candidates and then feeds the best observed class-score movement back to the generator for local refinement. The study asks: *(RQ1)* can score-guided refinement add valid label changes within the same six-query budget as one-round generation, and *(RQ2)* does any observed signal persist beyond deliberately selected low-margin nodes? Our contributions are:

- a topology-preserving score-only threat model that changes only the target-node text representation;
- a reproducible two-round search and validity pipeline with matched graph-aware and non-graph controls; and
- a two-stage evaluation showing that the apparent gain on a low-margin stress sample does not reproduce on an independent class-balanced random sample.

II. RELATED WORK

A. Graph Adversarial Robustness

Nettack demonstrated targeted evasion through constrained structure and feature perturbations [2]; Metattack extended this line to poisoning through meta-gradients [3]. Recent TAG work has shifted attention from abstract embeddings to interpretable language. *Intruding with Words* studies text-level graph injection and shows that textual interpretability changes the attack–utility trade-off [4]. GRAPHFEEDBACK differs by adding no node or edge: it replaces only an existing target's encoded text in a copied ego-graph.

B. Textual Black-Box Search

TextFooler ranks influential words and searches semantically similar replacements [5], while BERT-Attack uses masked-language-model candidates [6]. PromptAttack shows that an instruction-tuned model can generate its own evaluation inputs under explicit fidelity constraints [7]. Query-efficient attacks formalize the trade-off between attack success and victim interactions [8]. GRAPHFEEDBACK combines prompt-based generation with observed victim scores, but it does not estimate gradients or claim globally optimal edits.

III. THREAT MODEL AND METHOD

A. Score-Only Topology-Preserving Setting

Let $G_i = (V_i, E_i, X_i)$ be the copied ego-graph of an initially correct target node i, with original text x_i and class y_i. The victim returns a score vector $\mathbf{p}_i(x) \in \mathbb{R}^C$:

$$\hat{y}_i(x) = \arg \max_{c \in \{1,\ldots,C\}} p_{i,c}(x). \tag{1}$$

The evaluator observes these scores but not gradients, parameters, hidden representations, or training data. A candidate is encoded with the same all-MiniLM-L6-v2 family used by GraphCLIP [9]; only the copied root feature is replaced. Edges, neighbours, positional encodings, class prompts, and victim parameters remain fixed.

For an initially correct target, define the clean-class margin

$$m_i(x) = p_{i,y_i}(x) - \max_{c \neq y_i} p_{i,c}(x). \tag{2}$$

A successful valid edit makes $m_i(x) < 0$. If no queried candidate succeeds, the lowest-margin valid candidate supplies the refinement direction:

$$x_i^\star = \arg \min_{x \in \mathcal{C}_i^{\text{valid}}} m_i(x). \tag{3}$$

B. Two-Round Score-Guided Search

Figure 1 summarizes the procedure. Round 1 generates at most three candidates from the target text, class information, objective, and compact one-hop context. After filtering, candidates are evaluated sequentially with early stopping on success. Otherwise, Round 2 receives the current best candidate, its class scores, its margin and margin change, and a compact summary of first-round movements. It generates at most three local refinements:

$$\mathcal{C}_i = \mathcal{C}_i^{(1)} \cup \mathcal{C}_i^{(2)}, \quad |\mathcal{C}_i^{(1)}| \leq 3, \; |\mathcal{C}_i^{(2)}| \leq 3. \tag{4}$$

All second-round candidates are checked against the *original* text, preventing accumulated drift. The matched one-round graph baseline uses the identical first three candidates; the feedback comparison therefore isolates refinement rather than stochastic regeneration.

C. Candidate Validity

Before any victim query, a candidate must contain plain English text without explanatory wrappers, have a token-length ratio in $[0.8, 1.2]$, modify at most 20% of aligned tokens, and reach cosine similarity at least 0.85 under an independent all-mpnet-base-v2 Sentence-BERT encoder [10]. Numbers, citations, explicit negations, polarity terms, acronyms, and detected model names are preserved. Invalid or unparsable candidates consume generation effort but no victim query. These deterministic checks improve auditability but do not replace human meaning judgments.

IV. Experimental Methodology

A. Victim, Data, and Comparisons

We evaluate the released GraphCLIP checkpoint on CiteSeer [11]. Clean test accuracy is 69.28% (442/638). Qwen2.5-0.5B-Instruct [12] generates candidates at temperature 0.7 with at most 256 new tokens. Experiments use Python 3.10.20, PyTorch 2.4.1/CUDA 12.1, and an NVIDIA RTX 4050 Laptop GPU.

The first run, `stress30v1`, contains 30 initially correct lower-margin nodes, five per class. It evaluates random edit, generic paraphrase, one-round non-graph and graph prompts, and matched two-round feedback variants. The frozen `random60v1` confirmation selects 60 initially correct nodes at random within class, ten per class, excludes every stress identifier, and omits only generic paraphrasing to control runtime. Prompts, generator, query budget, stopping rules, and validity thresholds remain fixed. The samples are analyzed separately and never pooled.

B. Metrics and Statistical Treatment

For N initially correct targets, success requires both a valid edit and a changed prediction:

$$s_i = \mathbb{I}[x_i^\star \text{ valid} \land \hat{y}_i(x_i^\star) \neq y_i], \quad \text{ASR} = \frac{1}{N} \sum_{i=1}^{N} s_i. \tag{5}$$

Nodes with no valid candidate remain failures in the denominator. Candidate reliability is

$$\text{NVCR} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\big[|\mathcal{C}_i^{\text{valid}}| = 0\big]. \tag{6}$$

We also report mean victim queries, median semantic similarity, aligned changed-token ratio, and margin reduction

$$\Delta m_i = m_i(x_i) - m_i(x_i^\star). \tag{7}$$

ASR intervals use 10,000 seeded node-bootstrap resamples. Because methods attack the same nodes within each run, comparisons use paired binary outcomes and a two-sided exact McNemar-style test. With few successes and discordant pairs, these tests are descriptive and low-powered.

V. Results

A. Low-Margin Stress Evaluation

Table I shows that GraphFeedback changes 7/30 stress targets, while its matched graph prompt changes 5/30. Five successes are already present among the shared first-round candidates; feedback adds two. The paired outcome contains two GraphFeedback-only and zero graph-prompt-only successes ($p = 0.5$). The direction is consistent with useful refinement on this fixed sample, but neither the wide bootstrap interval nor the exact test supports superiority. Feedback without graph context also reaches 7/30 and adds three refinement-only successes. Against this control, the two feedback methods each have five unique successes ($p = 1.0$): graph context changes *which* nodes are affected, not aggregate ASR. Random edit reaches 5/30, confirming that the stress sample is broadly sensitive to small perturbations.

B. Frozen Confirmatory Evaluation

The confirmatory run reverses the optimistic stress interpretation. GraphFeedback and graph prompt both achieve 0/60; non-graph feedback achieves 2/60 and its one-round baseline 1/60. GraphFeedback has no unique success against non-graph feedback, which has two ($p = 0.5$). Figure 3 shows that

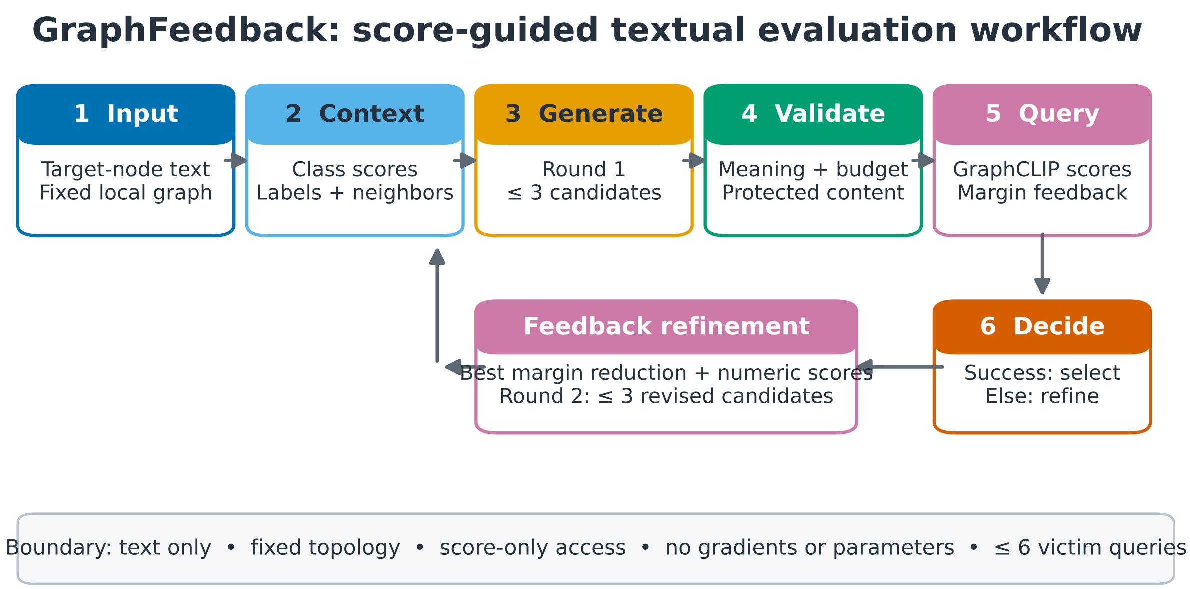

Fig. 1. The GRAPHFEEDBACK pipeline. The generator proposes constrained textual candidates, validity filtering occurs before victim evaluation, and observed GraphCLIP scores guide one bounded refinement round. Only the copied target-node feature changes.

TABLE I

RESULTS ON THE LOW-MARGIN STRESS SAMPLE AND INDEPENDENT CLASS-BALANCED RANDOM SAMPLE. "FB" DENOTES FEEDBACK. MEAN Q IS THE MEAN NUMBER OF VALID CANDIDATES SUBMITTED TO GRAPHCLIP. STRESS AND RANDOM SAMPLES ARE DISJOINT AND ARE NOT POOLED.

Run	Method	Success	ASR % (95% CI)	Mean Q	NVCR %	Median sim.
Stress ($N = 30$)	Random edit	5/30	16.7 [3.3, 30.0]	5.63	0.0	0.9979
	Generic paraphrase	5/30	16.7 [3.3, 30.0]	1.80	13.3	0.9875
	Non-graph	4/30	13.3 [3.3, 26.7]	1.80	13.3	0.9924
	Graph prompt	5/30	16.7 [3.3, 30.0]	1.90	6.7	0.9921
	Non-graph FB	7/30	23.3 [10.0, 40.0]	2.77	3.3	0.9873
	GRAPHFEEDBACK	7/30	23.3 [10.0, 40.0]	2.50	0.0	0.9896
Random ($N = 60$)	Random edit	0/60	0.0 [0.0, 0.0]	5.73	0.0	0.9983
	Non-graph	1/60	1.7 [0.0, 5.0]	2.02	15.0	0.9888
	Graph prompt	0/60	0.0 [0.0, 0.0]	1.53	28.3	0.9935
	Non-graph FB	2/60	3.3 [0.0, 8.3]	2.87	3.3	0.9863
	GRAPHFEEDBACK	0/60	0.0 [0.0, 0.0]	2.27	15.0	0.9918

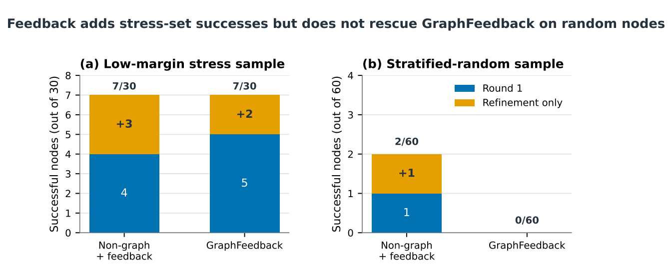

Fig. 2. First-round and refinement-only successes. Feedback adds stress-set successes, but the graph-aware effect does not persist in confirmation.

random edit and both graph-aware methods decline together from the stress sample to zero. This coordinated decline is more consistent with target difficulty than with a method-specific failure. The stress run establishes feasibility close to selected decision boundaries; the random run bounds generalization.

C. Efficiency, Reliability, and Case Analysis

GRAPHFEEDBACK uses 2.50 victim queries per stress node and 2.27 per random node, below the six-query maximum because success triggers early stopping and invalid text is not queried. In the random run, the one-round graph prompt produces no valid candidate for 17/60 nodes, compared with 9/60 for GRAPHFEEDBACK. Feedback therefore improves candidate availability, but availability alone does not imply a boundary crossing.

Stress node 1544 illustrates a refinement-only success. Three valid first-round candidates preserve the Artificial Intelligence prediction. The fourth query, a feedback-guided refinement, changes it to Machine Learning; the clean-class margin moves from 0.9972 to -0.3558, sentence similarity is 0.9653, and

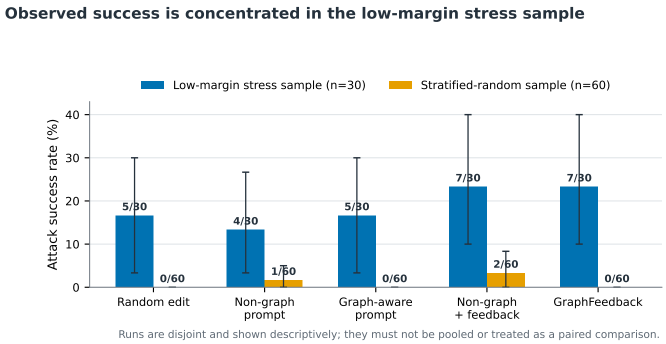

Fig. 3. ASR under lower-margin stress sampling and class-balanced random sampling. The disjoint samples are neither paired nor pooled.

4.85% of aligned tokens change. The random run contains no GRAPHFEEDBACK success. Its only refinement-only case occurs under non-graph feedback and includes debatable semantic shifts despite similarity 0.9429. These cases show why embedding similarity cannot establish human-equivalent meaning.

VI. Discussion

The two runs distinguish operational feasibility from broad effectiveness. Closed-loop score feedback can improve local search near selected low-margin targets without exceeding the one-round method's maximum query budget. However, the frozen random sample does not support a general graph-aware advantage. The most defensible finding is *margin sensitivity*: target selection materially changes apparent vulnerability.

Several limitations constrain the claim. Only one CiteSeer checkpoint, one small local generator, and two finite samples are evaluated. Successes are sparse, making intervals wide and paired tests low-powered. Semantic fidelity is automatic rather than blindly human-rated. The confirmatory seed also changes upstream stochastic graph construction, so it is an independent replication rather than an identical-universe resample. Future work should test additional TAG datasets, checkpoints, generators, seeds, and query budgets, and should include blinded semantic and factual annotation.

VII. Conclusion

GRAPHFEEDBACK is a reproducible score-guided framework for topology-preserving textual robustness evaluation of GraphCLIP. It identifies two refinement-only successes beyond a matched one-round graph-aware baseline on a 30-node lower-margin stress sample, reaching 7/30 total successes. The result does not persist on a non-overlapping 60-node class-balanced random sample, where GRAPHFEEDBACK reaches 0/60. Accordingly, the work does not establish universal vulnerability or method superiority. It demonstrates that bounded victim feedback can guide local textual search while showing that representative robustness conclusions depend critically on sampling and semantic validation.

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